



IILM University

GREATER NOIDA · U.P.

School of Computer Science
& Engineering

INTERNSHIP PRESENTATION · 2026

WEB DEVELOPMENT INTERNSHIP

ShopEase

A Responsive E-Commerce Website

A one-month front-end development internship — building a multi-page e-commerce front-end with category filters, keyword search, persistent cart, and a working contact form.

STUDENT

Dhananjay Parashar

PROGRAM

B.Tech CSE

3rd Semester · Batch 2025-2029

ORGANIZATION

Codec Technologies Pvt. Ltd.

Web Developer Intern · 01/08/2026 -- 30/08/2026

Where I spent the summer.

A one-month, AICTE & ICAC approved Web Developer Internship run through the National Internship Portal — hosted by an ed-tech & skill-development organization that partners with Google for Education.

01 • ORGANIZATION

Codec Technologies Pvt. Ltd.

An ed-tech & skill-development organization that partners with the National Internship Portal (an initiative under the Ministry of Education) to deliver AICTE & ICAC approved internship programs to engineering and technology students across India.

In association with Google for Education, Codec conducts structured, mentor-guided internship programs in Web Development, Data Science, Machine Learning and Cybersecurity.

4 domains

Web • Data Science • ML • Cyber

AICTE

& ICAC approved program

Google

For Education Partner

02 • THE PROGRAM

Web Developer Internship

ROLE

Web Developer Intern

DURATION

1 Month

FOCUS

Move students from theoretical classroom knowledge to practical, project-based learning through real development workflows, coding best practices, and modern front-end frameworks.

What I built — and why.

PROBLEM STATEMENT

Small and first-time online sellers struggle to showcase products without investing in a heavy, expensive e-commerce platform. Beginner templates are not responsive, lack a working cart, or do not allow users to search and filter efficiently.

The need: a lightweight, responsive e-commerce front-end that lets visitors browse, search, filter, and add items to a cart — with cart data persisting across reloads, without a heavy back-end.

01 • PROJECT OBJECTIVES

Six concrete goals.

- 01 Design a clean, responsive, multi-page website across desktop, tablet, mobile.
- 02 Dynamic product listing with category-based filtering & keyword search.
- 03 Functional cart: add, update quantity, remove items.
- 04 Persist cart data across page reloads using browser `localStorage`.
- 05 Client-side validation of a working contact / enquiry form.
- 06 Semantic HTML, good coding practices, Git/GitHub version control.

02 • SCOPE

Front-end, intentionally.

Scope is limited to the front-end of an e-commerce website — covering a home page, product listing with filter & search, product detail view, cart page, and contact page.

Static / local JSON is used to simulate the catalogue rather than connecting to a live back-end or payment gateway, since the focus was front-end development, responsive design, and client-side logic.

Real back-end, database & payment gateway are flagged as natural extensions beyond the internship.

Stack & system architecture.

A simple client-side, layered architecture — built entirely on industry-standard front-end tools, versioned through Git/GitHub.

01 • TECH STACK

Major technologies & tools.

CATEGORY	TECHNOLOGY	PURPOSE
Markup & Styling	HTML5, CSS3, Bootstrap 5	Page structure & responsive UI
Scripting	JavaScript (ES6)	Cart, search, form validation
Data	JSON	Local product catalogue
Storage	localStorage	Persist cart across sessions
Version Control	Git & GitHub	Source-code management
Tools	VS Code, Chrome DevTools	Development & testing

Table 4.1 — Technologies and Tools Used

02 • SYSTEM ARCHITECTURE

Four layers, client-side.

```
graph TD; User[User (Browser)] --- P1[ ]; P1 --- PL[Presentation Layer<br/>HTML5 + CSS3 + Bootstrap 5<br/>Responsive UI]; P1 --- AL[Application Logic<br/>JavaScript (ES6)<br/>Cart, Filter, Validation]; PL --> AL; P2[ ]; P2 --- DL[Data Layer<br/>JSON Product Catalogue]; P2 --- BS[Browser Storage<br/>localStorage (Cart Persistence)];
```

The diagram illustrates the client-side system architecture of ShopEase. At the top is the 'User (Browser)'. Below it, a horizontal line branches into two paths. The left path leads to the 'Presentation Layer' (HTML5 + CSS3 + Bootstrap 5, Responsive UI). The right path leads to 'Application Logic' (JavaScript (ES6), Cart, Filter, Validation). A horizontal arrow points from the Presentation Layer to the Application Logic. Below these two layers, another horizontal line branches into two paths. The left path leads to the 'Data Layer' (JSON Product Catalogue). The right path leads to 'Browser Storage' (localStorage (Cart Persistence)).

Figure 4.1 — System Architecture of ShopEase

METHODOLOGY

Requirements → Design → Development → Testing → Refinement & Deployment

What I take away.

Five concrete deliverables, hands-on proficiency in industry-standard tools, and a structured workflow that survives the next project.

01

Fully responsive, multi-page front-end.

04

Improved proficiency in HTML5, CSS3, Bootstrap, JS (ES6).

02

Cart with add / update / remove, persisted via localStorage.

05

Hands-on Git/GitHub version control & structured dev workflow.

03

Searchable & filterable product listing.

Thank you.

Open to questions,
feedback & discussion.

